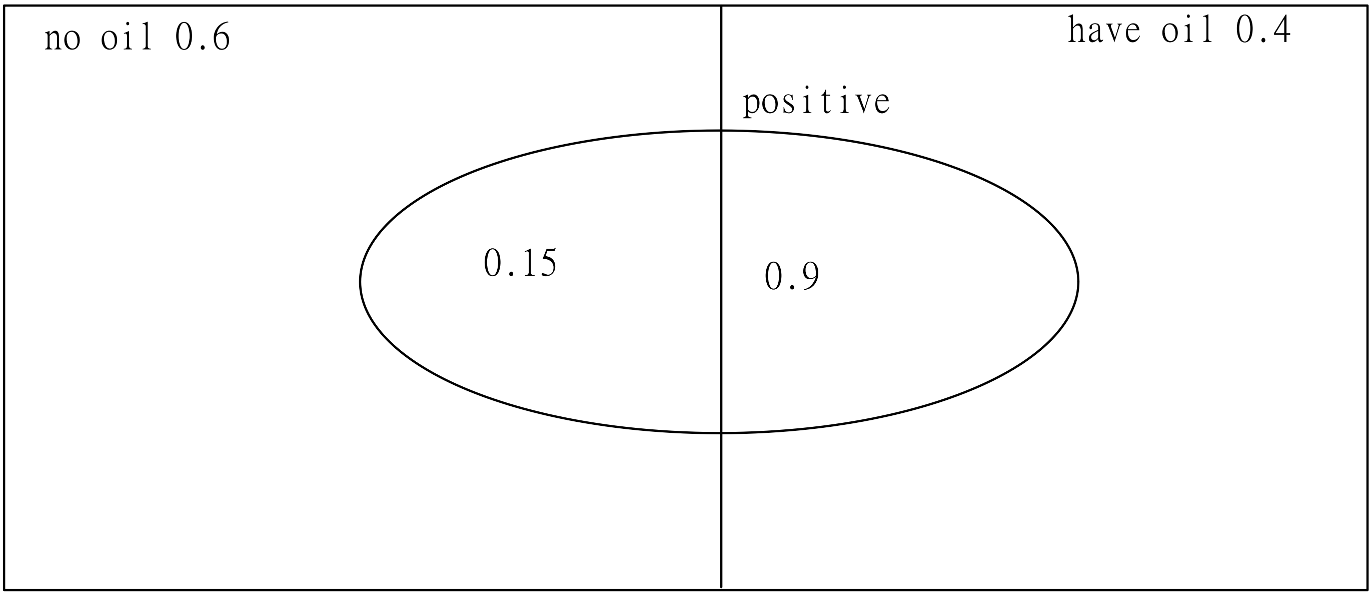
•An
oil explorer performs a seismic test to determine whether oil is likely to be found in a certain area. The probability that the test indicates the presence of oil is 90% if oil is indeed present in the test area and the probability of a false positive is 15% if no oil is present in the test area. Before the test is done, the explorer believes that the probability of presence of oil in the test area is 40%. Use Bayes' rule to revise the value of the probability of oil being present in the test area given that the test gives a positive signal.



P(have oil | positive) = P(have oil \inter positive)/P(positive)

P(positive | have oil) = P(have oil \inter positive)/P(have oil)
0.9 = P(have oil \inter positive)/0.4
0.36 = P(have oil \inter positive)

P(have oil | positive) = 0.36/P(positive)

P(positive) = P(positive \inter have oil) + P(positive \inter no oil)
P(positive) = P(positive | have oil)P(oil) + P(positive | no oil)P(no oil)
P(positive) = (0.9)(0.4) + (0.15)(0.6)
P(positive) = 0.45

P(have oil | positive) = 0.36/0.45
P(have oil | positive) = 0.8